

EE605 Quiz 2 Solutions

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1 Question 1

The $(n, 1)$ repetition code C comprises the n -length codewords $\{0 \cdots 0, 1 \cdots 1\}$ with minimum distance $d = n$. This code can correct a maximum of $t = \lfloor \frac{d-1}{2} \rfloor$ errors. A binary code is perfect if the Hamming balls of radius t around the codewords are disjoint and cover the entire space $\{0, 1\}^n$, i.e. the code satisfies the Hamming bound with equality.

Therefore, for C to be a perfect code, $(\# \text{codewords in } C)(\# \text{binary words in a Hamming ball of radius } t) = 2^n$

$$2 \sum_{i=0}^t \binom{n}{i} = 2^n \implies \sum_{i=0}^t \binom{n}{i} = 2^{n-1}$$

Consider the following two cases.

1. n is odd : Then $n = 2k + 1 \implies t = k$, and

$$\sum_{i=0}^k \binom{2k+1}{i} = \sum_{i=0}^k \binom{2k+1}{2k+1-i} = \sum_{i=k+1}^{2k+1} \binom{2k+1}{i}$$

Thus,

$$\sum_{i=0}^k \binom{2k+1}{i} = \sum_{i=0}^k \binom{2k+1}{2k+1-i} = \frac{2^{2k+1}}{2} = 2^{2k} = 2^{n-1}$$

The code is therefore a perfect code for odd n .

2. n is even : Then $n = 2k \implies t = k - 1$, and

$$\sum_{i=0}^{k-1} \binom{2k}{i} = \sum_{i=0}^{k-1} \binom{2k}{2k-i} = \sum_{i=k+1}^{2k} \binom{2k}{i}$$

Thus,

$$\sum_{i=0}^{k-1} \binom{2k}{i} = \sum_{i=0}^{k-1} \binom{2k}{2k-i} = \frac{2^{2k} - \binom{2k}{k}}{2} < 2^{2k-1} = 2^{n-1}$$

The code is therefore **not** a perfect code for even n .

2 Question 2

2.a

The given function cannot be one-to-many, as a single codeword cannot have multiple distinct prefixes of length $n - d + 1$.

Now, assume that the function is many-to-one, i.e. $\exists c, \bar{c} \in C$ s.t. $g(c) = g(\bar{c})$. This means that codewords c and $\bar{c}$ are identical in the first $n - d + 1$ coordinates, and hence can differ only in the last $d - 1$ coordinates. But this is a contradiction since $d_H(c, \bar{c}) \leq d - 1 < d$. Therefore, the function is not many-to-one and is **one-to-one**.

2.b

Proceed by arranging the m n -length codewords of C as rows of an $m \times n$ matrix. d is the minimum distance of the code. The total pairwise Hamming distance between all pairs of codewords of C is given by

$$\sum_i \sum_j d(c_i, c_j) = \sum_{i=1}^n \sum_{j=0}^{q-1} x_i^{(j)} (m - x_i^{(j)})$$

where $x_i^{(j)}$ is the number of occurrences of symbol $j \in \text{GF}(q)$ in column i .

The LHS is lower bounded by $m(m-1)d$. The RHS is maximized when $x_i^{(j)} = \frac{m}{q}$, $\forall j$. Hence,

$$m(m-1)d \leq nm^2 \frac{q-1}{q}$$

This gives

$$m \leq \frac{qd}{qd - (q-1)n}$$

3 Question 3

3.a

An (n, k) linear code C is MDS iff it has minimum distance $d = n - k + 1$, or equivalently the minimum codeword weight is $n - k + 1$, which implies that the number of zero coordinates in a non-zero codeword of C is at most $k - 1$.

Let there exist a set $I = \{i_1, \dots, i_k\}$ of k columns which are linearly dependent. Then there exists a non-zero message $u \in \mathbb{F}_q^k$ s.t. $uG_I = \{u \cdot g_{i_1}, \dots, u \cdot g_{i_k}\} = 0$, which implies that the number of zeros in the codeword uG is at least k . Thus, $w_H(uG) \leq n - k$. But this contradicts the minimum distance $d = n - k + 1$ of an MDS code.

Hence, for an MDS code C , any subset of k columns of the generator matrix G must be linearly independent.

3.b

Code C is MDS $\iff d = n - k + 1$. The dual code $C^\perp$ of C is an $[n, n - k, d^\perp]$ linear code. For an $[n, n - k, d^\perp]$ linear code, the Singleton bound gives $d^\perp \leq k + 1$. For $C^\perp$ to be an MDS code, it must satisfy the singleton bound with equality, i.e. $d^\perp = k + 1$. Therefore we need to prove $d^\perp \geq k + 1$ for $C^\perp$ to be an MDS code.

Let G be the generator matrix of C . From the previous result, any subset of k columns of G are linearly independent. Now, G is also the parity check matrix of $C^\perp$. Therefore, the minimum Hamming weight of any non-zero codeword of $C^\perp$ must be at least $k + 1$, which implies $d^\perp \geq k + 1$.

Therefore, the dual code $C^\perp$ of an MDS code C is also an MDS code.

4 Question 4

4.a

The order of $G = [G_1 | G_2]$ is clearly $k \times (n_1 + n_2)$. First we need to show that G is full row-rank (otherwise it is not a generator matrix of a linear code). Indeed, say $a_1 g_1 + \dots + a_k g_k = 0$ where g_i is the i^{th} row of G . Then in particular the first n_1 components of $a_1 g_1 + \dots + a_k g_k$ are 0 i.e. $a_1 g_{11} + \dots + a_k g_{1k} = 0$ where g_{1i} denotes the i^{th} row of G_1 . Since G_1 is full row-rank, $a_1 = \dots = a_k = 0$. So G is full row-rank. Now let C be the linear code generated by G . That the length of C is $n_1 + n_2$ and that $\dim(C) = k$ are obvious from the order of G .

The only non-trivial part is finding the minimum distance of C . We will only be able to bound it from below here. Let C_1, C_2 be the linear codes generated by G_1, G_2 respectively. See that the codewords of C can be formed by a linear combination of the rows of G . So the tuple of the first n_1 components of any codeword of C is a linear combination of the rows of G_1 i.e. a codeword of C_1 . More precisely,

$$C \subset \{(c_1 | c_2) : c_1 \in C_1, c_2 \in C_2\}$$

For any codeword $u = (u_1 | u_2)$ of C , $\text{wt}(u) = \text{wt}(u_1) + \text{wt}(u_2)$. The key observation is to notice that $u = 0 \iff u_1 = 0$ and $u_2 = 0$. To see this, see that if $u = 0$, then so are its first n_1 components, and so the coefficients of

the linear combination of the rows of G which yielded u must be 0 (G_1 is full row-rank). So all the components of u are 0. Now, for nonzero u , $\text{wt}(u) = \text{wt}(u_1) + \text{wt}(u_2) \geq d_1 + d_2$. So the best lower bound we can obtain is $d(C) \geq d_1 + d_2$. It is also easy to construct examples where this lower bound is achieved.

4.b

We can similarly show that

$$G = \begin{pmatrix} G_1 & 0 \\ 0 & G_2 \end{pmatrix}$$

is full row-rank. It has order $2k \times (n_1 + n_2)$. We show that

$$C = \{(c_1 | c_2) : c_1 \in C_1, c_2 \in C_2\}$$

where C is the code generated by G . For every $c \in C$, there exist unique coefficients $a_1, a_2, \dots, a_{2k}$ such that $c = a_1 g_1 + \dots + a_{2k} g_{2k}$. Then $c = (c_1 | c_2)$, where $c_1 = a_1 g_{11} + \dots + a_k g_{1k} \in C_1$ and $c_2 = a_{k+1} g_{21} + \dots + a_{2k} g_{2k} \in C_2$. Conversely, it is easy to see that $(c_1 | c_2) \in C$ by just choosing the first k coefficients as those of c_1 and the last k as those of c_2 (when expressed as a linear combination of rows of G_1, G_2 respectively).

So the length and dimension of C are $n_1 + n_2$ and $2k$ respectively. Further, the minimum distance of any codeword of C will be equal to $\min(\{d_1, d_2\})$. To show this, let $u = (u_1 | u_2) \in C$ be nonzero. Then $\text{wt}(u) = \text{wt}(u_1) + \text{wt}(u_2) < \min(\{d_1, d_2\}) \Rightarrow \text{wt}(u_1) < d_1$ and $\text{wt}(u_2) < d_2 \Rightarrow u_1 = 0$ and $u_2 = 0 \Rightarrow u = 0$, which is a contradiction. And of course, WLOG, if $d_1 = \min(\{d_1, d_2\})$, then the codeword $(u_1 | 0)$ of C has weight equal to $d_1 = \min(\{d_1, d_2\})$, where u_1 is a minimum weight codeword of C_1 .

5 Question 5

5.a

Let C be an $[n, k]$ -code, and let $P \subseteq \{1, \dots, n\}$. Then the shortening of C at P is defined as

$$C_P = \{x^P : x \in C \text{ and } x_i = 0 \forall i \in P\}$$

where x^P is the tuple of components of the codeword x with indices not in P . So we have

$$RM(1, m)_{\{1\}} = \{(x_2, \dots, x_{2^m}) : x \in RM(1, m) \text{ and } x_1 = 0\}$$

Note that $x \in RM(1, m) \Rightarrow x = ag + a_1 g_1 + \dots + a_m g_m$ (where $a, a_i \in \mathbb{F}_2$, g_i is the i^{th} row of G_1 , and g is the “all-ones” element of $\mathbb{F}_2^m$). Then the first component of x is simply a since the first column of G_1 is $0 \in \mathbb{F}_2^m$. Thus, $x_1 = 0 \Rightarrow a = 0 \Rightarrow x$ is a linear combination of the rows of G_1 . Further, we are anyways removing the first coordinate of x . Now write

$$G = \begin{pmatrix} 1 \\ G_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & G'_1 \end{pmatrix}$$

Then we see that every element of C' is a linear combination of the rows of G'_1 . So G'_1 is a generator matrix for C' . Notice that G'_1 is an $m \times (2^m - 1)$ matrix consisting of all the nonzero vectors in $\mathbb{F}_2^m$ as its columns.

5.b

The length and dimension of C' can be directly read off from the order of G'_1 i.e. $2^m - 1$ and m respectively. The minimum distance needs some work. Say $u \in C' \setminus \{0\}$. Then u is the sum of some k rows from the rows of G'_1 , where $1 \leq k \leq m$. There are 2^k distinct elements of $\mathbb{F}_2^k$. Let us see how many times these elements occur among the columns of this $k \times (2^m - 1)$ submatrix of G'_1 (rows of which add up to u). Clearly, the number of times $v \in \mathbb{F}_2^k$ occurs in this manner is equal to the number of elements of $\mathbb{F}_2^m \setminus \{0\}$ which contain v as a prefix. This is clearly 2^{m-k} for nonzero v . $0 \in \mathbb{F}_2^k$ occurs only $2^{m-k} - 1$ times since we have discarded the all-zero vector. Now, $\text{wt}(u) =$ number of ‘1’s in u . Of all $v \in \mathbb{F}_2^k$, exactly 2^{k-1} have an odd number of ‘1’s and 2^{k-1} have an even number of ‘1’s. Thus, the number of ‘1’s in $u =$ number of times an element with an odd number of ‘1’s occurs in the columns of the $k \times (2^m - 1)$ submatrix $= 2^{k-1} \times 2^{m-k} = 2^{m-1}$. We have shown that every nonzero codeword of C' has weight 2^{m-1} . So in particular, $d(C') = 2^{m-1}$.

5.c

The binary Hamming code $\mathcal{H}_m(2)$ has a parity check matrix of it as an $m \times \left(\frac{2^m-1}{2-1}\right) = m \times (2^m - 1)$ matrix consisting of all the nonzero vectors of $\mathbb{F}_2^m$. The code C' has this as a generator matrix. So $C' = \mathcal{H}_m(2)^\perp = \mathcal{S}_m(2)$ i.e. C' is a binary simplex code of order m .